INTEGRATION

BLAKE FARMAN

Lafayette College

Name: Solutions

Use right endpoints and the formulas

- $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$

to evaluate the following integrals.

1.

$$\int_0^3 x^3 \, \mathrm{d}x.$$

$$\Delta x = \frac{3-0}{n} = 3/n, \quad x_i = 0 + i\Delta x = i\left(\frac{3}{n}\right) = \frac{3i}{n}$$

$$f(x_i)\Delta x = \left(\frac{3i}{n}\right)^3\left(\frac{3}{n}\right) = \frac{27i^3}{n^3}\left(\frac{3}{n}\right) = \frac{81i^3}{n^4}$$

$$\int_0^3 x^3 dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{81i^3}{n^4} = \lim_{n\to\infty} \frac{81}{n^4}\sum i^3$$

$$= \lim_{n\to\infty} \frac{81}{n^4}\left(\frac{n(n+1)}{2}\right)^2 = \lim_{n\to\infty} \frac{81}{4}\left(\frac{n^4+2n^3+n^2}{n^4}\right)$$

$$= \lim_{n\to\infty} \frac{81}{4}\left(\frac{1+2/n+1/n^2}{1}\right) = \frac{81}{4}\left(\frac{1+0+0}{1}\right) = \boxed{\frac{81}{4}}$$

2.

$$\int_1^2 x^2 \, \mathrm{d}x$$

$$\Delta x = \frac{2-1}{n} = \frac{1}{n}, \quad x_i = 1 + i\Delta x = 1 + \frac{i}{n} = \frac{n+i}{n}$$

$$f(x_i)\Delta x = \left(\frac{n+i}{n}\right)^2\left(\frac{1}{n}\right) = \frac{n^2+2in+i^2}{n^3} = \frac{1}{n} + \frac{2i}{n^2} + \frac{i^2}{n^3}$$

$$\int_1^2 x^2 dx = \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{1}{n} + \frac{2i}{n^2} + \frac{i^2}{n^3}\right) = \lim_{n\to\infty} \left[\frac{1}{n}\sum_{i=1}^{n} 1 + \frac{2}{n^2}\sum_{i=1}^{n} i + \frac{1}{n^3}\sum_{i=1}^{n} i^3\right]$$

$$= \lim_{n\to\infty} \left[\frac{1}{n}(n) + \frac{2}{n^2}\left(\frac{n(n+1)}{2}\right) + \frac{1}{n^3}\left(\frac{n(n+1)(2n+1)}{6}\right)\right]$$

$$= \lim_{n\to\infty} \left[1 + \frac{n+1}{n} + \frac{2n^2+3n+1}{6n^2}\right] = \lim_{n\to\infty} \left[1 + 1 + \frac{1}{n} + \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}\right]$$

$$= 1 + 1 + 0 + \frac{1}{3} + 0 + 0 = 2 + \frac{1}{3} = \boxed{\frac{7}{3}}$$